Cyprus University of Technology



Computer Engineering & Informatics

CEI122: Ψηφιακή Λογική

CEI_122 - Report Εργαστηρίου 8

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08/04/2025

Quartus II 64-Bit - U:/2nd Semester/Psifiaki_logiki/team_n_lab8_8_4_2025/ex1 - ex1

File Edit View Project Assignments Processing Tools Window Help

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Project Navigator

Entity

Cyclone II: EP2C20F484C7

ex1

Program:

ex1.v | ex1_tb.v | Compilation Report - ex1

```
1  module ex1(
2      input clk,
3      input reset_n,
4      input w,
5      output reg z,
6      output [8:0] state
7  );
8
9
10 parameter State_A = 9'b000000001;
11 parameter State_B = 9'b000000010;
12 parameter State_C = 9'b000000100;
13 parameter State_D = 9'b000001000;
14 parameter State_E = 9'b000010000;
15 parameter State_F = 9'b000100000;
16 parameter State_G = 9'b001000000;
17 parameter State_H = 9'b010000000;
18 parameter State_I = 9'b100000000;
19
20 reg [8:0] current_state, next_state;
21
22
23 always @(posedge clk or negedge reset_n) begin
24     if (!reset_n)
25         current_state <= State_A;
26     else
27         current_state <= next_state;
28 end
29
30
31 always @(*) begin
32     case (current_state)
33         State_A: next_state = (w == 1) ? State_B : State_F;
34         State_B: next_state = (w == 1) ? State_C : State_A;
35         State_C: next_state = (w == 1) ? State_D : State_A;
36         State_D: next_state = (w == 1) ? State_E : State_A;
37         State_E: next_state = (w == 1) ? State_E : State_A;
38         State_F: next_state = (w == 0) ? State_G : State_A;
39         State_G: next_state = (w == 0) ? State_H : State_A;
```

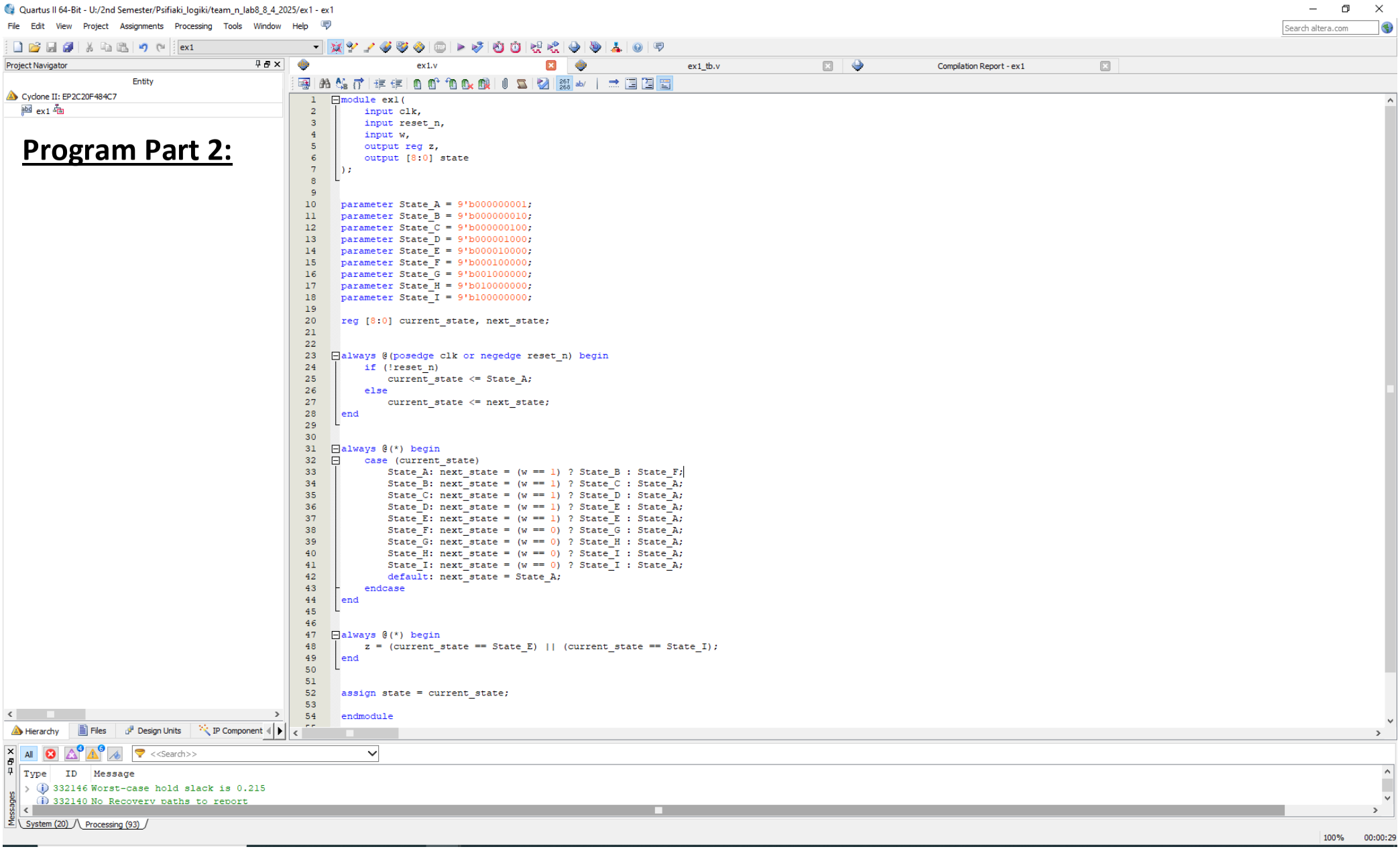
Hierarchy | Files | Design Units | IP Component

All | <<Search>>

Type	ID	Message
	332146	Worst-case hold slack is 0.215
	332140	No Recovery paths to report
	332140	No Removal paths to report
	332148	Timing requirements not met
	332146	Worst-case minimum pulse width slack is -1.222
	332001	The selected device family is not supported by the report_metastability command.
	332102	Design is not fully constrained for setup requirements
	332102	Design is not fully constrained for hold requirements
		Quartus II 64-Bit TimeQuest Timing Analyzer was successful. 0 errors, 4 warnings
		**********
		Running Quartus II 64-Bit EDA Netlist Writer
		Command: quartus_eda --read_settings_files=off --write_settings_files=off ex1 -c ex1
	204026	Generated files "ex1.vo", "ex1_fast.vo", "ex1_v.sdo" and "ex1_v_fast.sdo" in directory "U:/2nd Semester/Psifiaki_logiki/team_n_lab8_8_4_2025/simulation/modelsim/" for EDA simulation tool
		Quartus II 64-Bit EDA Netlist Writer was successful. 0 errors, 0 warnings
	293000	Quartus II Full Compilation was successful. 0 errors, 10 warnings

Messages: System (20) | Processing (93)

100% 00:00:29



Program Part 2:

```
module ex1(
    input clk,
    input reset_n,
    input w,
    output reg z,
    output [8:0] state
);


parameter State_A = 9'b000000001;
parameter State_B = 9'b000000010;
parameter State_C = 9'b000000100;
parameter State_D = 9'b000001000;
parameter State_E = 9'b000010000;
parameter State_F = 9'b000100000;
parameter State_G = 9'b001000000;
parameter State_H = 9'b010000000;
parameter State_I = 9'b100000000;

reg [8:0] current_state, next_state;

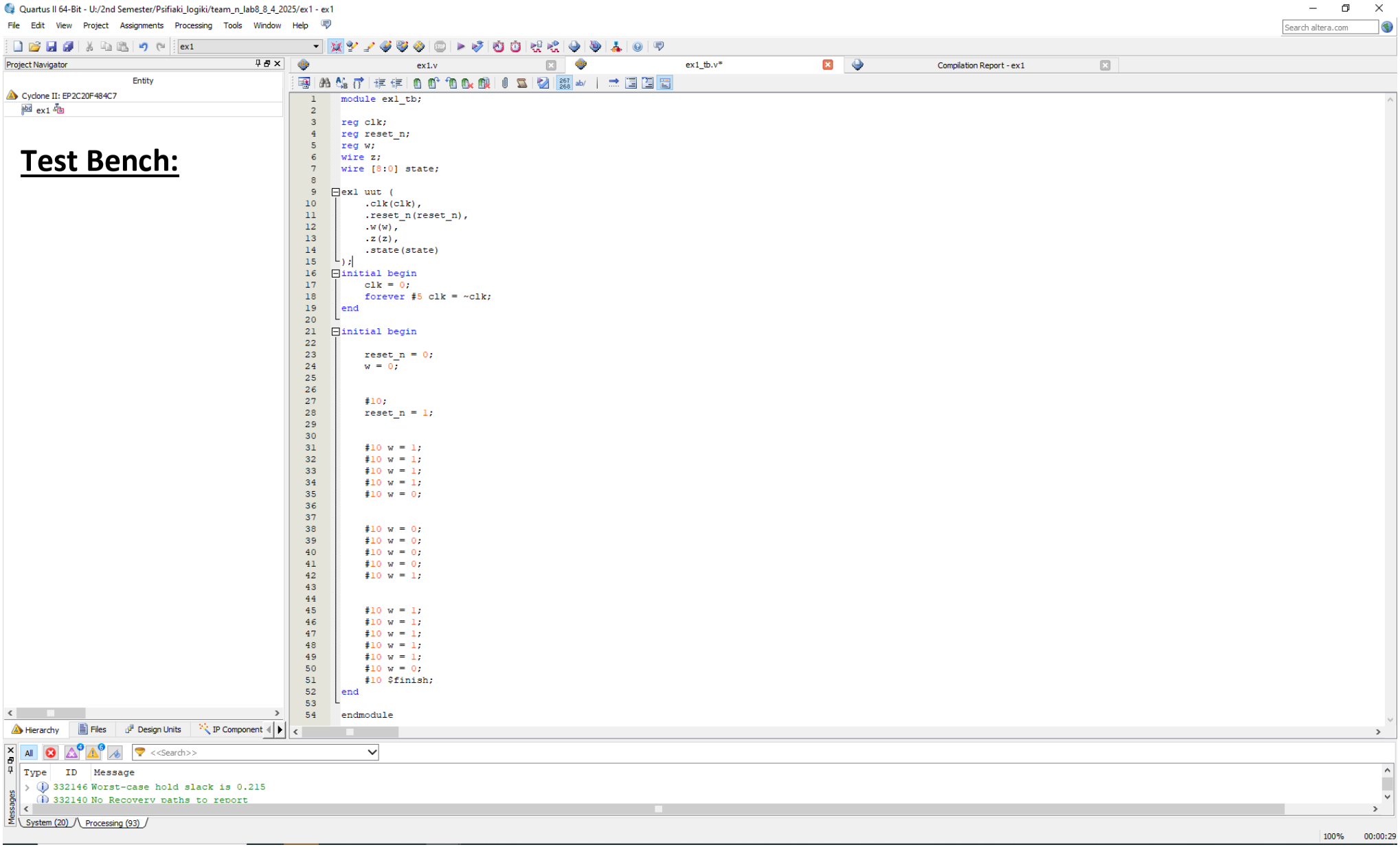

always @(posedge clk or negedge reset_n) begin
    if (!reset_n)
        current_state <= State_A;
    else
        current_state <= next_state;
end


always @(*) begin
    case (current_state)
        State_A: next_state = (w == 1) ? State_B : State_F;
        State_B: next_state = (w == 1) ? State_C : State_A;
        State_C: next_state = (w == 1) ? State_D : State_A;
        State_D: next_state = (w == 1) ? State_E : State_A;
        State_E: next_state = (w == 1) ? State_E : State_A;
        State_F: next_state = (w == 0) ? State_G : State_A;
        State_G: next_state = (w == 0) ? State_H : State_A;
        State_H: next_state = (w == 0) ? State_I : State_A;
        State_I: next_state = (w == 0) ? State_I : State_A;
        default: next_state = State_A;
    endcase
end

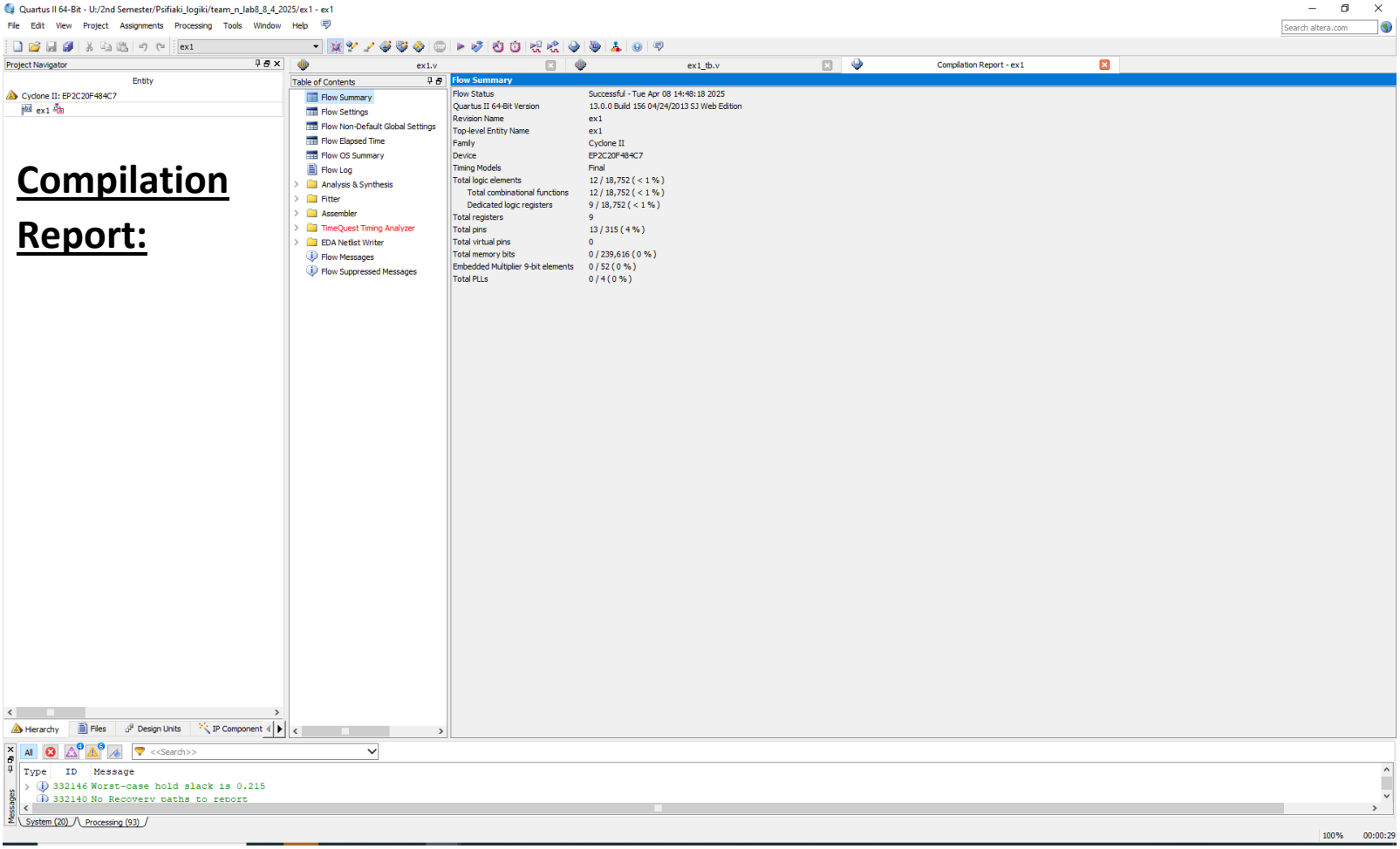

always @(*) begin
    z = (current_state == State_E) || (current_state == State_I);
end


assign state = current_state;

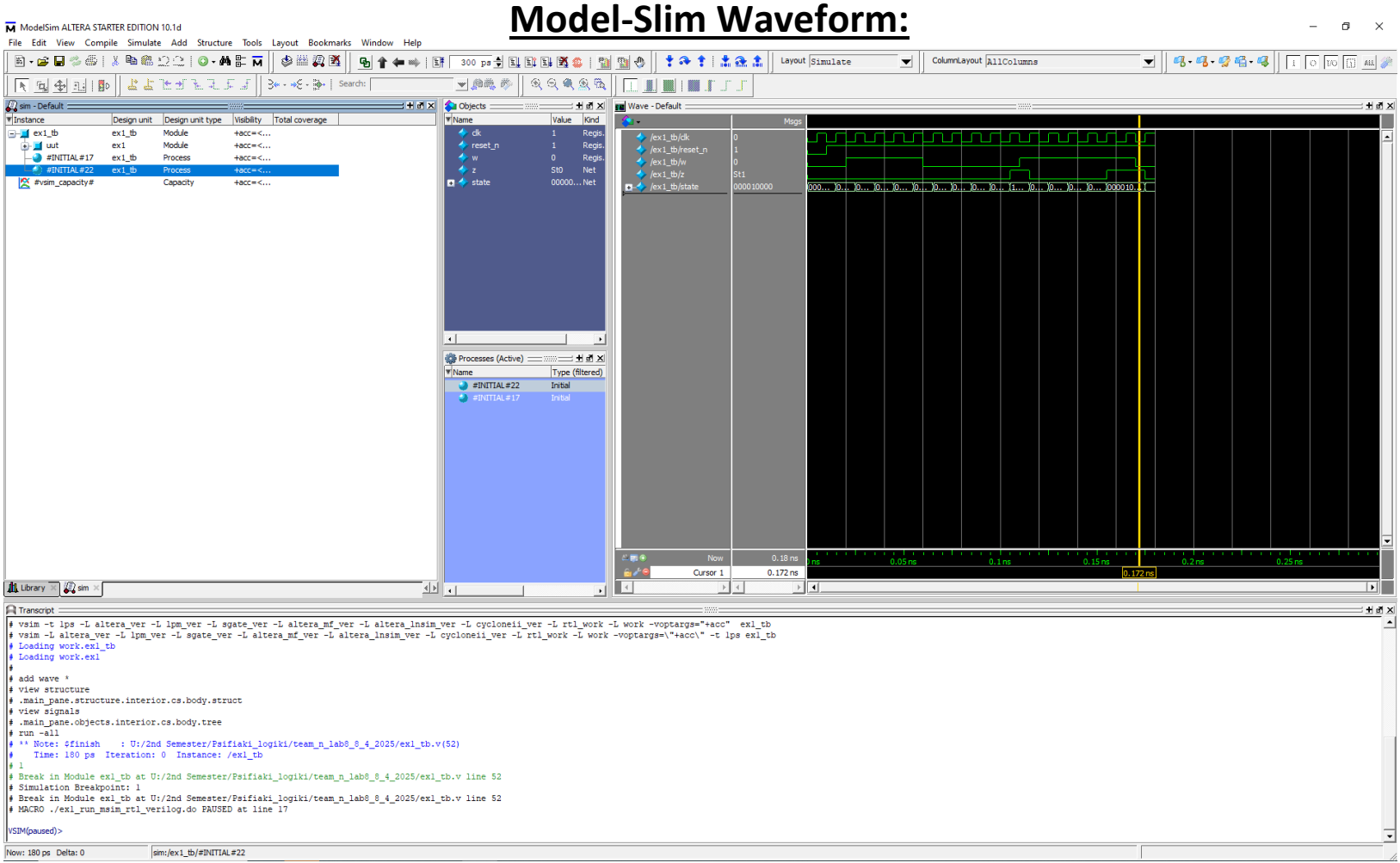
endmodule
```


Quartus II 64-Bit - U:/2nd Semester/Psifiaki_logiki/team_n_lab8_8_4_2025/ex1 - ex1
File Edit View Project Assignments Processing Tools Window Help
Search altera.com
Project Navigator
Entity
Cyclone II: EP2C20F484C7
ex1
ex1.v
ex1_tb.v*
Compilation Report - ex1
Test Bench:
1 module ex1_tb;
2
3 reg clk;
4 reg reset_n;
5 reg w;
6 wire z;
7 wire [8:0] state;
8
9 ex1 uut (
10 .clk(clk),
11 .reset_n(reset_n),
12 .w(w),
13 .z(z),
14 .state(state)
15);
16 initial begin
17 clk = 0;
18 forever #5 clk = ~clk;
19 end
20
21 initial begin
22
23 reset_n = 0;
24 w = 0;
25
26
27 #10;
28 reset_n = 1;
29
30
31 #10 w = 1;
32 #10 w = 1;
33 #10 w = 1;
34 #10 w = 1;
35 #10 w = 0;
36
37
38 #10 w = 0;
39 #10 w = 0;
40 #10 w = 0;
41 #10 w = 0;
42 #10 w = 1;
43
44
45 #10 w = 1;
46 #10 w = 1;
47 #10 w = 1;
48 #10 w = 1;
49 #10 w = 1;
50 #10 w = 0;
51 #10 $finish;
52 end
53
54 endmodule
Hierarchy
Files
Design Units
IP Component
<<Search>>
Type ID Message
332146 Worst-case hold slack is 0.215
332140 No Recovery paths to report
System (20)
Processing (93)
Messages
100%
00:00:29

Compilation Report:



Model-Slim Waveform:



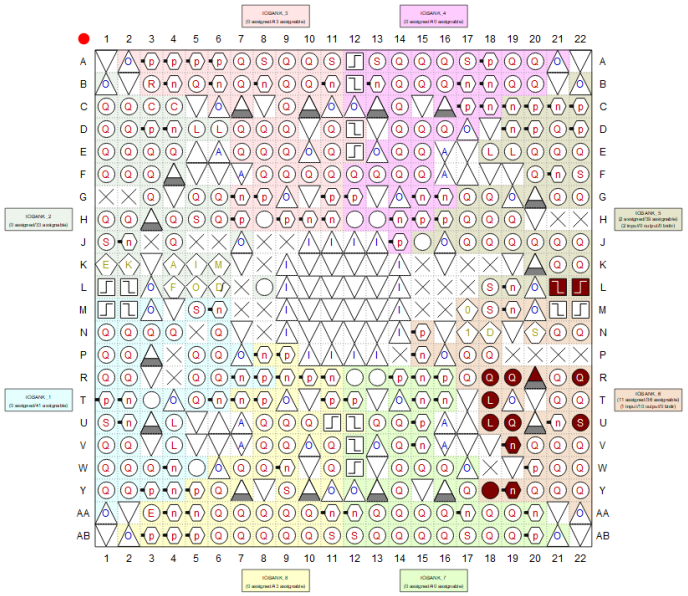
Pin Planner - U:/2nd Semester/Psifiaki_logiki/team_n_lab8_8_4_2025/ex1 - ex1

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Pin Planner:

Top View - Wire Bond
Cyclone II - EP2C20F484C7



Groups

Named: *

Node Name	Direction
state[8..0]	Output Group
<<new group>>	

Groups Report

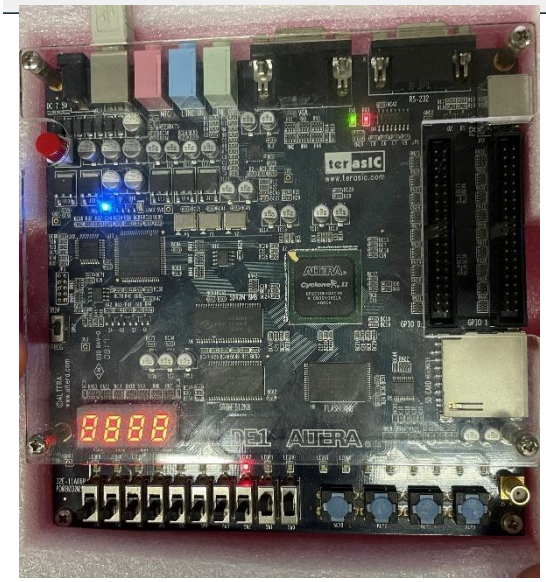
Tasks

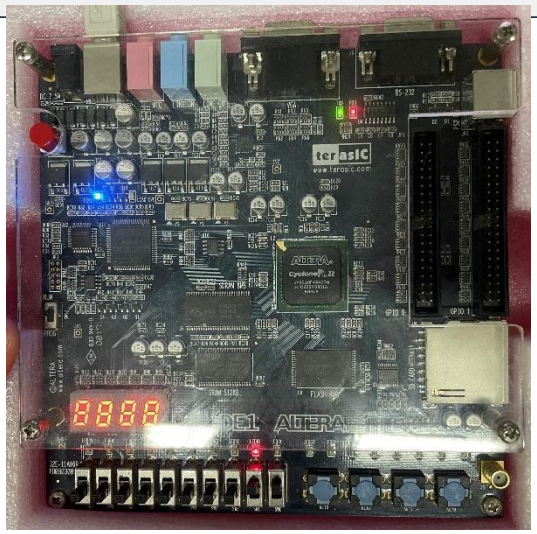
- Run Analysis and Elaboration
- Early Pin Planning
  - Early Pin Planning...
  - Run I/O Assignment Analysis
  - Export Pin Assignments...
- Change View
  - Show I/O Banks
  - Show VREF Groups
  - Show Edges
  - Show DQ/DQS Pins
    - x4 Mode
    - x8 Mode
    - x16 Mode
    - x32 Mode
  - Show Clock Region Input Pi
    - Global Clock Region
    - Quadrant Clock Region
    - Periphery Clock Region
    - LVDS Clock Region
    - Spine Clock Region

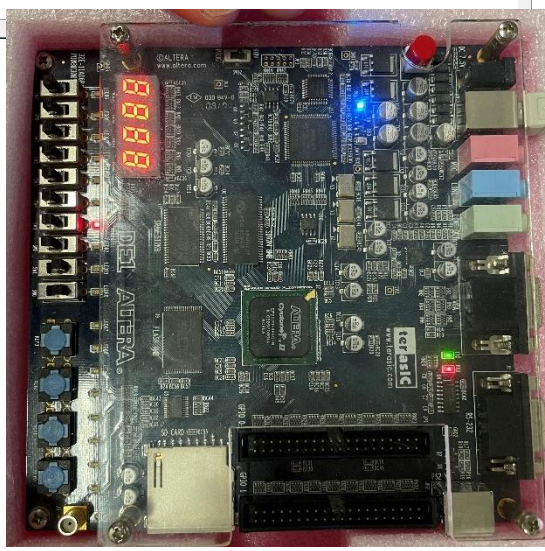
Named: * Edit: Filter: Pins: all

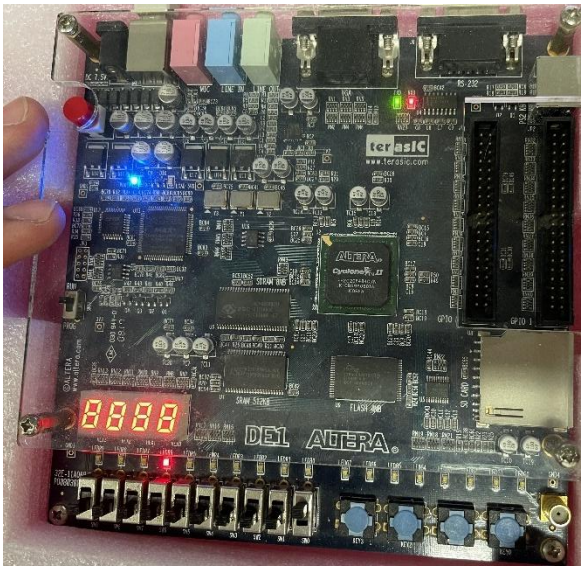
Node Name	Direction	Location	I/O Bank	VREF Group	Fitter Location	I/O Standard	Reserved	Current Strength	Differential Pair
clk	Input	PIN_R22	6	B6_N0	PIN_R22	3.3-V LV...default)		24mA (default)	
reset_n	Input	PIN_L22	5	B5_N1	PIN_L22	3.3-V LV...default)		24mA (default)	
state[8]	Output	PIN_R18	6	B6_N0	PIN_R18	3.3-V LV...default)		24mA (default)	
state[7]	Output	PIN_U18	6	B6_N1	PIN_U18	3.3-V LV...default)		24mA (default)	
state[6]	Output	PIN_Y18	6	B6_N1	PIN_Y18	3.3-V LV...default)		24mA (default)	
state[5]	Output	PIN_V19	6	B6_N1	PIN_V19	3.3-V LV...default)		24mA (default)	
state[4]	Output	PIN_T18	6	B6_N1	PIN_T18	3.3-V LV...default)		24mA (default)	
state[3]	Output	PIN_Y19	6	B6_N1	PIN_Y19	3.3-V LV...default)		24mA (default)	
state[2]	Output	PIN_U19	6	B6_N1	PIN_U19	3.3-V LV...default)		24mA (default)	
state[1]	Output	PIN_R19	6	B6_N0	PIN_R19	3.3-V LV...default)		24mA (default)	
state[0]	Output	PIN_R20	6	B6_N0	PIN_R20	3.3-V LV...default)		24mA (default)	
w	Input	PIN_L21	5	B5_N1	PIN_L21	3.3-V LV...default)		24mA (default)	

All Pins









Programmer - U:/2nd Semester/Psifiaki_logiki/team_n_lab8_8_4_2025/ex1 - ex1 - [Chain2.cdf]*

File Edit View Processing Tools Window Help

Altera Programmer:

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Hardware Setup... USB-Blaster [USB-0] Mode: JTAG Progress: 100% (Successful)

Enable real-time ISP to allow background programming (for MAX II and MAX V devices)

Start
Stop
Auto Detect
Delete
Add File...
Change File...
Save File
Add Device...
Up
Down

File	Device	Checksum	Usercode	Program/ Configure	Verify	Blank-Check	Examine
output_files/ex1.sof	EP2C20F484	001B2B8A	001B2B8A	☑	☐	☐	☐

